# Auditing Anonymized Tick Data: A Bias-Robust Framework for Detecting Affine Scaling and Directional Inversion in Intraday Markets


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**Abstract**

When quantitative firms share proprietary tick data with external auditors under NDA, they routinely apply sanitizing transformations — principally affine price scaling ($P' = \alpha P + \beta$) and directional inversion ($P' = -P + C$) — to prevent alpha leakage. Standard audit pipelines, calibrated for exchange-native tick sizes and expecting the classical negative leverage effect, fail silently under both transformations. We make three contributions. First, we show that the tick size of an affine-scaled series can be recovered exactly via the greatest common divisor (GCD) of absolute price movements, bypassing the scaling constant entirely. Second, we apply the leverage-effect puzzle of Aït-Sahalia, Fan and Li (2011) to the inversion-detection problem, demonstrating that at 1-minute frequency the standard lagged correlation estimator produces near-zero or reversed signals on *all* markets — not merely on Indian equity as sometimes claimed — making leverage-based detectors unreliable. Third, we show that the order-book bid-ask constraint (*Ask* > *Bid*) is violated in 100% of rows after inversion when Level-2 data is available, providing a definitive, bias-free test. We validate all three results on 22,419 one-minute bars across 12 NSE large-cap symbols and release the detection framework as open auditing infrastructure.



**Keywords:** data masking · tick-data audit · leverage effect puzzle · GCD tick recovery · bid-ask impossibility · NSE intraday microstructure · Engle-Ng sign bias


## 1 Introduction

The independent auditing of proprietary algorithmic trading pipelines requires access to the underlying price series. Firms sharing such data under non-disclosure agreements face a genuine conflict: the time series itself carries information about strategy, alpha, and execution quality beyond the structural defects the auditor is meant to assess. The practical resolution — widespread but under-documented — is to supply the auditor with a sanitized transform of the data rather than the raw feed. Two transformations are particularly common: *affine price scaling*, where all prices are multiplied by an unknown constant and shifted ($P' = \alpha P + \beta$), and *directional inversion*, where the sign of all returns is reversed ($P' = -P + C$ for some large constant $C$ that keeps prices positive).

Standard audit tooling fails silently under both transformations. Affine scaling breaks tick-size-based exchange identification and any check calibrated to absolute price levels. Directional inversion turns a profitable long strategy into an apparently unprofitable short one and vice versa — an auditor unaware of the inversion reaches structurally incorrect conclusions about data quality even while all individual checks pass. The problem is not that the defects are hidden; it is that the audit engine's reference frame is wrong.

We address both failure modes with three contributions that together form a coherent detection framework requiring no knowledge of the scaling constants and no reference dataset.

**Contribution 1 (GCD tick recovery).** Because all valid prices are integer multiples of the fundamental tick size, the tick size of an affine-scaled series equals the GCD of all non-zero absolute price movements — regardless of the scaling constant $\alpha$. We prove this formally and show that the estimator is computable in $O(n)$ time, is robust to microstructure noise through a percentile filter, and identifies the exchange (NSE, US markets) with greater than 80% confidence when combined with session-bar counting.

**Contribution 2 (Leverage-puzzle implication).** Detection of directional inversion has been proposed via the leverage effect: after inversion, the classical negative return-volatility correlation should turn positive. We apply the formal results of Aït-Sahalia, Fan and Li (2011) to show that at one-minute sampling frequency this estimator is subject to severe discretization, smoothing, and estimation biases that drive the measured correlation toward zero or positive values on *all assets*, regardless of whether inversion has occurred. We confirm this empirically on NSE intraday data: mean $L(1) = +0.010$ across 12 symbols, consistent with the puzzle rather than with genuine reversed leverage. Standard leverage-based detectors are therefore unreliable at intraday frequency.

**Contribution 3 (L2 bid-ask impossibility).** When Level-2 order-book data is available alongside the price series, inversion can be detected definitively. Market mechanics require $Ask > Bid$ at all times. After directional inversion with any additive constant, the bid and ask columns exchange roles: $Ask' < Bid'$ in 100% of rows. This is a structural impossibility that no amount of noise can mask. We formalize the proof and show that it is robust to partial inversion (violation rate > 5%) and independent of market context.

The remainder of the paper is organized as follows. Section 2 presents the GCD tick recovery method. Section 3 reviews the leverage effect puzzle and its implications for inversion detection. Section 4 describes the full detection framework. Section 5 presents empirical results on NSE intraday data. Section 6 concludes.

## 2 GCD-Based Recovery of the Fundamental Tick Size

### *2.1 The affine scaling problem*

Let $\{P_t\}$ denote a price series with fundamental tick size $\tau$, so that all prices lie on the lattice $\tau \cdot \mathbb{N}$. After affine scaling, the auditor observes $P'_t = \alpha P_t + \beta$ for unknown constants $\alpha > 0$ and $\beta$. The tick size of the transformed series is $\alpha\tau$, which is not a member of any standard exchange tick set and renders all exchange-identification heuristics based on minimum price increments void. A detector using the minimum non-zero price movement as a tick estimate recovers $\alpha\tau$ rather than $\tau$ and classifies the exchange incorrectly.

### *2.2 Recovery via greatest common divisor*

Let $\Delta_i = |P'_i - P'_{i-1}|$ denote the sequence of absolute price movements. Since all prices lie on $\alpha\tau \cdot \mathbb{N}$, every non-zero $\Delta_i$ is an integer multiple of $\alpha\tau$. The GCD of any set of integer multiples of a common unit equals that unit (up to integer factors):

$$GCD(\Delta_1, \Delta_2, \ldots, \Delta_n) = \alpha\tau \cdot GCD(k_1, k_2, \ldots, k_n) \quad (1)$$

where $k_i = \Delta_i / (\alpha\tau) \in \mathbb{N}$. Since the set of integers $\{k_i\}$ almost surely contains 1 (achieved whenever a one-tick price move occurs) and 2, their GCD is 1. Therefore:

$$GCD(\{\Delta_i : \Delta_i > 0\}) = \alpha\tau \quad (2)$$

This result is exact and independent of $\beta$. The recovered quantity $\alpha\tau$ is compared against the set of known tick sizes for each exchange: for NSE, $\tau \in \{0.05, 0.10\}$; for US equity, $\tau = 0.01$. If $\alpha\tau$ matches no known tick, affine scaling is flagged. If it does match, the exchange is identified and $\alpha$ is recovered.

### *2.3 Practical implementation*

In practice, floating-point arithmetic introduces rounding error. We scale all movements by 100,000 before computing integer GCD, which handles tick sizes up to five decimal places without loss. A minimum-sample guard of 10 non-zero movements is applied. The estimator runs in O($n$) time. Validated on simulated EGARCH price series: the estimator recovers the correct tick size exactly in all cases where at least one one-tick movement is present in the sample, which holds with probability exceeding 99% for $n \geq 200$ bars.

## 3 The Leverage Effect Puzzle and Its Implications

### *3.1 Background*

The leverage effect — the negative correlation between an asset's return and its subsequent volatility — has been proposed as a basis for detecting directional inversion. The intuition is straightforward: after inversion, what were previously negative returns (causing high future volatility) become positive returns, reversing the correlation. Formally, if the pre-inversion series satisfies $Corr(r_t, \sigma^2_{t+1}) < 0$, then the inverted series satisfies $Corr(-r_t, \sigma^2_{t+1}) > 0$, providing a directional test.

### *3.2 The puzzle at high frequency*

Aït-Sahalia, Fan and Li (2011) demonstrate that at high sampling frequency the natural estimator of this correlation

$$\hat{\rho} = Corr(\hat{V}_{t+\Delta,\Delta} - \hat{V}_{t,\Delta}, X_{t+\Delta} - X_t) \quad (3)$$

is subject to four compounding biases: a discretization bias, a smoothing bias from estimating integrated rather than spot volatility, an estimation-error shrinkage bias, and a microstructure noise correction bias. These biases are concave functions of the horizon parameter $m$ and, at $m = 1$ (the natural contemporaneous choice), drive $\hat{\rho}$ toward zero for *all* assets regardless of the true correlation. Table 5 of that paper shows TSRV-based estimates of +0.087 and PAV-based estimates of +0.030 for Microsoft at one-minute frequency — positive, despite well-documented negative daily leverage. The authors conclude that "the leverage effect cannot be detected empirically using a natural approach" at intraday frequency.

### 3.3 *Implications for inversion detection*

The puzzle creates a fundamental identification problem for leverage-based inversion detectors. Consider an auditor computing the lagged leverage cross-correlation $L(\tau) = Corr(r_t, r^2_{t+\tau})$ for $\tau = 1, \ldots, 5$ on a one-minute series. On clean data, the puzzle drives $L(1)$ toward zero or positive. On inverted data, the identity $L'(\tau) = -L(\tau)$ still holds mathematically — but if the pre-inversion $L(1)$ is already near zero due to estimation bias, the post-inversion value $-L(1)$ is also near zero and the detector cannot distinguish the two states. This is not a calibration failure specific to any market; it is a consequence of the latency of the volatility process at intraday frequency.

## 4 The Detection Framework

We organize detection into three tiers by decreasing reliability, each applicable under different data availability conditions.

### 4.1 *Tier 1 — L2 bid-ask impossibility (definitive)*

**Theorem.** Let $\{B_t, A_t\}$ denote a bid-ask pair with $A_t > B_t > 0$ (standard market structure). After directional inversion $P' = -P + C$, the transformed pair satisfies $A'_t < B'_t$.

*Proof.* $B'_t = -B_t + C$ and $A'_t = -A_t + C$. Since $A_t > B_t$, multiplying by −1 reverses the inequality: $-A_t < -B_t$, and adding the constant $C$ preserves the direction: $A'_t < B'_t$. □

Any additive constant $C$ used to keep transformed prices positive does not affect the inequality. A violation rate exceeding 5% is used as the detection threshold, yielding a confidence of 0.95 that is independent of market, asset class, or sampling frequency. When Level-2 data is present, this test is applied first and overrides all heuristic signals.

### 4.2 *Tier 2 — Engle-Ng sign bias (supplementary, puzzle-aware)*

When Level-2 data is unavailable, we apply the Engle-Ng (1993) sign bias test as a supplementary heuristic. The OLS regression

$$r^2_t = \alpha_0 + \alpha_1 \cdot S^-_{t-1} + \varepsilon_t \tag{4}$$

where $S^-_{t-1} = 1$ if $r_{t-1} < 0$, tests whether negative past shocks predict higher current squared returns. For western equity at daily frequency, the canonical result is $\alpha_1 > 0$. After inversion, the sign reverses. However, consistent with the leverage puzzle, this signal is unreliable at one-minute frequency. We apply it only as corroborating evidence and clearly flag its confidence cap at 0.80.

### 4.3 *Tier 3 — Market identification and affine detection*

Market context — required to interpret the sign of Tier-2 signals — is recovered from the GCD tick estimator (Section 2) combined with modal session bar count. NSE is identified by $\tau \in \{0.05, 0.10\}$ and 375 one-minute bars per session; US equity by $\tau = 0.01$ and 390 bars. When neither criterion is met, Tier-2 signals are flagged as ambiguous and only Tier 1 is reported.

The framework returns a structured result with confidence, method, and explicit evidence strings. When the market context is unknown and no Level-2 data is present, the framework reports `ambiguous=True` rather than making an unsupported claim.

## 5 Empirical Evidence: NSE Intraday 1-Minute Data

### *5.1 Data and setup*

We apply the leverage-signal diagnostic to a proprietary store of cleaned NSE one-minute bars for 12 large-cap symbols (ISINs available on request) covering the 60 most recent trading days available in the store, yielding 22,419 bars per symbol. Data are collected via the Upstox WebSocket feed, cleaned through a purpose-built pipeline (PIT store, session-aware gap-flagging, OHLC integrity checks), and stored in Hive-partitioned Parquet format. For each symbol we compute $L(1)$, the cumulative lagged leverage $\Sigma\tau=1,\ldots,5\ L(\tau)$, the Engle-Ng $t$-statistic, and the semi-variance ratio.

### *5.2 Results*

| Symbol (ISIN) | N | L(1) | Cum-L | Eng-t | SemiVar | Signal |
|---|---|---|---|---|---|---|
| INE002A01018 | 22,419 | +0.025 | +0.109 | −0.10 | 0.904 | Reversed |
| INE009A01021 | 22,419 | +0.001 | +0.012 | +0.34 | 0.922 | Reversed |
| INE040A01034 | 22,419 | +0.009 | +0.020 | −1.72 | 1.306 | Reversed |
| INE062A01020 | 22,419 | +0.043 | +0.032 | −1.57 | 0.863 | Reversed |
| INE075A01022 | 22,419 | −0.000 | −0.011 | −1.08 | 1.213 | Normal |
| INE090A01021 | 22,419 | +0.042 | +0.059 | +0.31 | 0.787 | Reversed |
| INE237A01036 | 22,419 | +0.006 | +0.004 | −0.61 | 1.223 | Mixed |
| INE238A01034 | 22,419 | +0.009 | +0.002 | −0.17 | 1.206 | Mixed |
| INE296A01032 | 22,419 | +0.005 | +0.009 | −0.72 | 0.991 | Mixed |
| INE423A01024 | 22,419 | −0.012 | +0.040 | +0.76 | 0.811 | Reversed |
| INE467B01029 | 22,419 | +0.003 | +0.015 | −0.89 | 1.082 | Mixed |
| INE585B01010 | 22,419 | −0.009 | −0.001 | +0.52 | 0.934 | Mixed |
| **Mean** | — | **+0.010** | **+0.024** | **−0.41** | **1.020** | — |

*Table 1. Leverage signal diagnostics for 12 NSE large-cap symbols at one-minute frequency. L(1) = Corr($r_t$, $r^2_{t+1}$). Cum-L = Στ=1…5 L(τ). Eng-t = Engle-Ng OLS t-statistic. SemiVar = E[$r^2$|r<0] / E[$r^2$|r>0]. Signal: Reversed = majority of signals point positive; Normal = negative; Mixed = inconclusive.*

### *5.3 Interpretation*

The mean $L(1)$ across all 12 symbols is +0.010 — positive — with 6 symbols classified as reversed and 5 as mixed or normal. Crucially, no Engle-Ng $t$-statistic exceeds |1.5|, the minimum threshold for a meaningful signal. These results are consistent with the Aït-Sahalia puzzle: the expected signal direction cannot be reliably read from one-minute data even when the underlying population leverage effect is non-zero. Karmakar (2007) and multiple subsequent EGARCH studies confirm that NSE daily returns do exhibit negative leverage at lower frequencies. The intraday picture is dominated by estimation bias.

This has a direct practical implication: an audit engine applying a leverage-based inversion test to NSE one-minute data would classify approximately half of clean (uninverted) symbols as potentially inverted, and might classify genuinely inverted data as clean if the pre-inversion $L(1)$ happened to be positive. The Tier-1 L2 test is immune to all of these issues by construction.

## 6 Conclusion

We have shown that the two most common institutional data masking transformations — affine scaling and directional inversion — can be addressed by a tiered framework that avoids reliance on volatility estimation. The GCD tick estimator recovers the fundamental price increment exactly, regardless of the affine scaling constant. The L2 bid-ask impossibility proof detects inversion definitively and without bias when order-book data is available. The Engle-Ng sign bias test provides supplementary evidence when it is not, subject to the honest caveat that the leverage effect puzzle renders it unreliable at intraday sampling frequency.

The framework is implemented in Python as a standalone audit module. It is designed to run on client-provided data without requiring any reference dataset, any knowledge of the underlying asset, or any connection to live market data. The code, diagnostic scripts, and a synthetic benchmark dataset with 20 documented defect types are available on GitHub.

Future work will extend the framework to dollar-bar and event-bar series where the GCD estimator requires adaptation, and will investigate whether Aït-Sahalia's bias-correction procedure for the leverage cross-correlation can recover a reliable inversion signal at intraday frequency in the presence of NSE microstructure noise.

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**Data availability.** The detection framework and synthetic benchmark dataset are available at [GitHub URL, to be added on publication]. The NSE intraday data underlying Table 1 is proprietary and cannot be redistributed; the diagnostic script is provided in the repository. · **Conflicts of interest.** None declared. · **Acknowledgements.** The author thanks Anthropic's Claude for research assistance and code review.